

Shivansh Pratap Singh

Aerial Robotics · Robot Learning · Learning-Based Control

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RESEARCH INTERESTS

My research focuses on learning and control for aerial robots operating under uncertain and complex dynamics. I develop learning-based dynamics and world models that work alongside nonlinear and adaptive control, with applications spanning aerial manipulation, uncertain vehicle–arm dynamics, and perception-driven autonomous grasping.

EDUCATION

International Institute of Information Technology, Hyderabad (IIIT-H) **2024 – Present**
M.S. by Research, Electronics & Communication Engineering — Robotics & Control *Hyderabad, India*

- **CGPA 9.67/10** · Advisor: **Prof. Spandan Roy**, Robotics Research Center. Thesis direction: learning-based and adaptive control for aerial manipulation.

Bundelkhand Institute of Engineering & Technology (BIET), Jhansi **2019 – 2023**
B.Tech., Mechanical Engineering *Jhansi, India*

PUBLICATIONS

3 first / co-first-author papers. Author list order preserved; [†] denotes equal contribution.

[P1] AeroGrab: A Unified Framework for Aerial Grasping in Cluttered Environments

Shivansh P. Singh, Naveen Sudheer Nair, Samaksh Ujjawal, Sarthak Mishra, Soham Patil, Rishabh Dev Yadav, Spandan Roy

IEEE/RSJ Int. Conf. on Intelligent Robots and Systems (IROS), 2026 Accepted arXiv

Unified aerial-grasping framework coupling target understanding, active perception, 6-DoF grasp generation, motion planning and closed-loop manipulation for autonomous grasping in clutter.

[P2] Reinforcement Learning with Inner-Loop Dynamics Estimator for Aerial Manipulation under Uncertainty

Shivansh P. Singh, Samaksh Ujjawal, Ishita Chaudhary, V. R. Vasudevan, Rishabh Dev Yadav, Spandan Roy

IEEE Int. Conf. on Automation Science and Engineering (CASE), 2026 Accepted arXiv

Hierarchical whole-body RL with an estimator-based inner loop that compensates payload and vehicle–manipulator coupling uncertainty; validated on a 3-DoF aerial manipulator in hardware.

[P3] AERMANI-Diffusion: Regime-Conditioned Diffusion for Dynamics Learning in Aerial Manipulators

Samaksh Ujjawal[†], Shivansh P. Singh[†], Naveen Sudheer Nair, Rishabh Dev Yadav, Wei Pan, Spandan Roy

23rd IFAC World Congress (IFAC), 2026 Accepted arXiv

Regime-conditioned diffusion model for non-stationary aerial-manipulator dynamics, capturing abrupt payload and configuration changes to improve prediction and closed-loop tracking.

[P4] Learn Structure, Adapt on the Fly: Multi-Scale Residual Learning and Online Adaptation for Aerial Manipulators

Samaksh Ujjawal, Naveen Sudheer Nair, Shivansh P. Singh, Rishabh Dev Yadav, Wei Pan, Spandan Roy

IEEE/RSJ Int. Conf. on Intelligent Robots and Systems (IROS), 2026 Accepted arXiv

Multi-scale residual dynamics model with lightweight latent-space online adaptation for aerial manipulation under unseen payloads and operating conditions.

[P5] Impedance and Stability Targeted Adaptation for Aerial Manipulator with Unknown Coupling Dynamics

Amitabh Sharma, Saksham Gupta, Shivansh P. Singh, Rishabh Dev Yadav, Hongyu Song, Wei Pan, Spandan Roy, Simone Baldi

25th Int. Conf. on Control, Automation and Systems (ICCAS), 2025 Published DOI

Adaptive impedance-control framework for aerial manipulators with unknown vehicle–arm coupling, pairing stability analysis with hardware payload-catching experiments.

RESEARCH EXPERIENCE

Robotics Student Fellow — Autonomous Systems Lab (ASL)

ETH Zürich

Jul 2026 – Aug 2026

Zürich, Switzerland

- Developed **disturbance-aware latent world models** for a tilt-rotor omnidirectional aerial robot, targeting unmodeled aerodynamic interactions, actuator coupling and sim-to-real dynamics mismatch.
- Built a sim-to-real learning pipeline combining physics-based domain randomization with **JEPA-style latent dynamics representations**, including low-latency deployment and inference profiling on NVIDIA Jetson.

Research Assistant — Control Lab, Robotics Research Center

IIIT Hyderabad • Advisor: Prof. Spandan Roy

Apr 2024 – Present

Hyderabad, India

- Research learning-enabled and adaptive control for **aerial manipulators** under vehicle–arm coupling, payload variation, contact interaction and configuration-dependent uncertainty.
- Built ROS 2 simulation and hardware pipelines spanning whole-body reinforcement learning, residual dynamics learning, adaptive and impedance control, and autonomous aerial manipulation.
- Ran real-world aerial grasping and manipulation experiments involving payload transitions, high-dynamic maneuvers, visual perception and learned/model-based control.

Robotics Engineer

Arka Aerospace

May 2023 – Mar 2024

Hyderabad, India

- Developed UAV autonomy and navigation modules, including AprilTag-based visual servoing for precision landing at **2–3 cm terminal accuracy**.
- Implemented decentralized multi-UAV coordination and swarming algorithms for formation and spatial-coordination tasks.

SELECTED PROJECTS

Real-Time Quadrotor Control: RRT Planning + MPC Tracking

Autonomous Flight

- Built an RRT → trajectory → MPC autonomy stack for aggressive quadrotor flight, reaching **100 Hz closed-loop optimization** with constrained thrust/body-rate control via efficient solving and warm starts.

Autonomous Ball Catcher (“Robotic Dustbin”)

 GitHub

- Built a mobile robot that predicts the 3D landing point of a thrown ball from multi-camera triangulation, physics-based trajectory fitting and RANSAC, then intercepts it via 30 Hz target streaming and closed-loop drive control.

TECHNICAL SKILLS

Programming	Python, C++, MATLAB, Bash; Git, Linux
Robot Software	ROS 2, PX4, ArduPilot, MAVROS / micro-ROS, Gazebo, Isaac Lab
Robot Learning	PyTorch; reinforcement learning (PPO, SAC); world models & JEPA; diffusion models; temporal representation learning
Control & Estimation	MPC / MPPI, INDI, adaptive control, impedance control, system identification, state estimation
Perception	RGB-D and VIO sensing, multi-view geometry, 6-DoF grasp generation, visual servoing
Sim-to-Real	Domain randomization, physics simulation, learned residual dynamics, dynamics identification
Deployment	NVIDIA Jetson, CUDA / GPU inference, real-time onboard autonomy, aerial-robot hardware integration